

INNOVATIONS IN LIGHTING For Smart Cities

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Smart lighting technologies have expanded into smart cities, smart homes as well as agriculture, delivering major benefits

Global smart city solutions based on Low Power Wide Area (LPWA) networks can significantly reduce energy consumption, carbon emissions and operational complexities. Innovations in wireless connectivity contribute to lower costs and optimum maintenance processes, with consequent benefits to the environment and municipalities' budget. Now, streetlights too come with centrally-controlled and managed LED lights. Intensity of the lights can be increased or decreased as per needs. These smart lights can also be controlled through voice commands.

Control4 Smart Lighting features out-of-the-box voice control of lights with Amazon Alexa-enabled devices. Homes where KNX is the standard can start a music playlist with KNX keypad, adjust the heating, ventilation and air-conditioning (HVAC) system, dim or brighten the lights using a handheld remote or use voice to turn off the entire house at night from anywhere through a mobile device. The houses can be customised for automation to fit individual lifestyles.

Anuj Dhir, vice president and business head, Wipro Commercial Lighting, explains, "In modern workspaces, new technology features enable performance-tracking with

real-time updates on lighting system status." Sumit Joshi, vice chairman and managing director, Philips Lighting India, adds, "A manager can track occupancy patterns, changes in temperature, light levels and more, while employees can personalise the lighting around their desks."

Technologies for smart streetlighting

Cellular and LPWA connectivity technologies are used in smart streetlighting. LTE Cat M1/NB1-enabled streetlighting makes cities safer and more sustainable. LPWA-connected smart city use cases include remote monitoring and predictive maintenance of streetlight networks. City planners and municipalities no longer need to design and install proprietary gateways. Using public cellular infrastructure they can avoid allocating resources and time for the operation and maintenance of proprietary communication infrastructures.

According to Diego Grassi, senior manager, product strategy cellular - IoT market development, u-blox, their smart streetlighting remote monitoring platform consists of smart street luminaires and Cat M1, NB-IoT (Internet of Things) or 2G cloud analytics platform. The streetlight controller equipped

Fig. 1: SARA-R412M-enabled streetlight platform architecture and network topology (Credit: u-blox)

